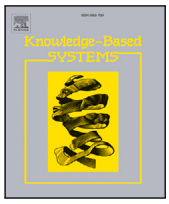
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Automated data synthesis and retrieval-augmented generation for legal large language models

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ABSTRACT

The application of large language models (LLMs) in the legal domain faces two critical challenges: data scarcity and the need for trustworthy outputs. Legal data is limited and often lacks the quality and structure necessary to train robust AI systems. Furthermore, legal tasks demand high accuracy and reliability, making it essential for AI systems to generate trustworthy content. To address these challenges, we present the first automated framework for generating high-quality retrieval-augmented generation (RAG) data tailored specifically for legal LLMs. Our approach involves synthesizing high-quality question–answer pairs relevant to the legal domain. Each generated pair is then evaluated by an LLM-as-a-Judge, which classifies them as either good or bad based on their quality. Subsequently, we apply an ELO-Ranking technique to rank all pairs, ensuring that the most reliable and accurate pairs are selected for further use. These high-quality pairs are then employed to train both the retrieval and generation modules of a RAG model, addressing the issue of data scarcity and improving the overall performance of legal LLMs. Experimental validation on the LawBench benchmark demonstrates that our approach significantly enhances model performance, providing a reliable solution for training trustworthy legal AI models with RAG-based techniques.

1. Introduction

With the rapid development of Artificial Intelligence (AI) and Natural Language Processing (NLP) technologies, large language models have demonstrated significant capabilities across various domains, including text generation, intelligent question answering, and document analysis. In the legal domain, the application of LLMs has shown great potential, particularly in automating the generation of legal documents, legal consultation, and case analysis. However, despite the promising results, the application of these technologies within the Chinese legal field faces several challenges, with one of the most critical being the scarcity of high-quality training data.

Legal data in China are relatively limited, especially standardized data and suitable for training AI systems. Legal resources in China consist mainly of statutes, judicial rulings, and policy documents, but these resources often have significant issues: first, most legal data lack proper structuring and standardization, making it difficult to process; second, many legal documents contain private or sensitive information, limiting their availability to the public; and third, the quantity and diversity of data remain insufficient to support large-scale model training.

This data scarcity directly hinders the development of AI-driven legal systems in China. For deep learning models, especially large language models, the quantity and quality of training data are crucial factors influencing their performance. This problem becomes particularly apparent in legal tasks, such as judicial reasoning and legal interpretation, where the lack of sufficient high-quality data prevents current models from achieving practical performance. Therefore, addressing the shortage of legal data, especially through data synthesis, is the key to enhance the effectiveness of legal AI systems.

As large language models have matured, many legal AI systems have begun to take advantage of their powerful text generation and comprehension abilities. However, their application in the legal domain faces unique challenges. Legal texts require models to generate precise and rigorous language, but large language models often fail to guarantee the accuracy and strictness required in legal documents. For example, models might generate common-sense errors when interpreting legal provisions or misinterpret the intent behind legal rulings. Additionally, the high level of abstraction in legal language may hinder models from

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